BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE

SESSION 2025-26
MPBA G512 — Time Series Analysis and Forecasting

Project Topic

Forecasting Inflation Using ARIMA and GARCH Models:

Evidence from IMF WEO Data Across Six Economies

Dataset: IMF World Economic Outlook, April 2025

Analysis Period: 1990–2024

SUBMITTED TO
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1. Introduction, Problem Statement, and Literature Review

1.1 Inflation and Its Macroeconomic Importance

Inflation — the sustained rise in the general price level — is among the most closely monitored macroeconomic indicators. Its dynamics directly influence central bank decisions, real household income, investment returns, and exchange rate stability. The IMF characterizes price stability as foundational to sustainable growth; persistent deviations impose wage-price spirals, erode purchasing power, and destabilize financial markets. Accurate inflation forecasting is therefore not merely academic: it is a practical requirement for monetary policy credibility, sovereign debt management, and corporate planning.

1.2 Problem Statement

This study asks: given only a country's own historical CPI inflation series, how well can a univariate time-series model predict future inflation? The question is operationalized using annual CPI percentage-change data for **six economies** — the United States (USA), Germany (DEU), Japan (JPN), the United Kingdom (GBR), China (CHN), and India (IND) — drawn from the IMF World Economic Outlook (WEO) database covering **1990 to 2024**. The Box-Jenkins ARIMA framework provides the primary forecasting engine, augmented by GARCH-class volatility models.

1.3 Literature Review

The Box-Jenkins methodology (Box and Jenkins, 1976) is the dominant framework for univariate time-series modelling and has been shown to be a competitive benchmark against structural models by Stockton and Glassman (1987) and Meyler, Kenny and Quinn (1998). The extension of Engle (1982) ARCH and Bollerslev (1986) GARCH introduced the time-varying volatility modelling for high-inflation series. Atkeson and Ohanian (2001) show how difficult it is to beat naive benchmarks without external information, a result we confirm here, while Ang, Bekaert and Wei (2007) find that ARIMAX and VAR models that incorporate macro drivers tend to dominate univariate specifications at medium horizons.

2. Methodology and Model Specification

2.1 Data Preprocessing

The IMF WEO dataset was limited to annual CPI percent change (1990-2024). A critical preprocessing step eliminated thousand-separator commas from hyperinflation observations (e.g. Brazil 1990: '2,947.73'), which R's as.numeric() converts to NA without warning. Missing values in the dataset (Russia 1990-1992; Nigeria 1990-1995) were imputed with zoo::na.approx preserving the Interpolated flag. Countries were grouped into four tiers for visualization based on the magnitude of inflation for within-group comparability on shared axes.

2.2 Stationarity Testing

Before model specification, each series underwent three complementary unit-root tests: the Augmented Dickey-Fuller (ADF) test and the Phillips-Perron (PP) test — both with a null of non-stationarity — and the KPSS test whose null is stationarity.

2.3 ARIMA Identification and Estimation

Model identification proceeded on two parallel tracks. First, auto.arima with stepwise search was applied to each training series. Second, an exhaustive grid search across $p \in \{0,1,2\}$, $d \in \{0,1\}$, $q \in \{0,1,2\}$ ranked each candidate by the bias-corrected AICc on the training set. Crucially, model selection (governed by AICc on training data) was separated from forecast evaluation (governed by RMSE on held-out data). The full-sample model (1990–2023) was used solely for the 2024–2026 point forecast.

2.4 Dual Evaluation Design

Two evaluation windows were constructed. The primary pre-COVID window trains on 1990–2014 and tests on 2015–2019 - a clean five-year period assessing ARIMA skill under normal macroeconomic conditions. The COVID-era stress test trains on 1990–2019 and tests on 2020–2023, deliberately spanning the pandemic shock and the 2021–2023 inflation surge. Because ARIMA cannot anticipate structural breaks, stress-test results are interpreted as a lower bound on model robustness rather than a measure of typical skill. The naive benchmark was specified as a random walk for all six countries, consistent with their ADF classification as I(1) series.

2.5 Forecast Accuracy Metrics

Four metrics were computed. RMSE (Root Mean Squared Error) is the primary ranking criterion, penalising large errors quadratically. MAE (Mean Absolute Error) provides an interpretable average absolute deviation in percentage points. sMAPE = $2|e|/(|actual|+|forecast|)$ replaces MAPE to avoid division by zero when inflation approaches zero — a material concern for Japan and China. The Skill Score = 1 − RMSE(ARIMA)/RMSE(Naive) quantifies percentage improvement over the random walk; a positive score means ARIMA adds value.

2.6 Volatility Modelling (ARCH/GARCH)

Following ARIMA estimation, squared residuals were tested for conditional heteroscedasticity via the ARCH-LM test at lags 5 and 10. Where evidence was found, GARCH(1,1) models under Gaussian and Student-t distributions were fitted. An EWMA ($\lambda = 0.94$) was also computed as a non-parametric volatility estimate. All GARCH results carry an explicit caveat: 33–34 annual observations fall well below the 100+ required for reliable GARCH estimation, and results should be interpreted as illustrative rather than definitive.

3. Data Description

The dataset comprises annual CPI inflation (percent change in average consumer prices) from the IMF WEO database, April 2025 edition, covering 196 countries from 1980 to 2024. The analytic window is restricted to 1990–2023 for modelling, with 2024 used as a single reference point against the IMF near-final estimate. Summary statistics for all fifteen examined economies are presented in Table 1.

Table 1: Descriptive Statistics — CPI Inflation (%, 1990–2023)

Country	N	Mean	Median	SD	Min	Max	Skew	CV%
United States	34	2.69	2.63	1.51	-0.32	7.99	1.20	56.0
United Kingdom	34	2.75	2.33	2.09	0.04	9.07	1.67	75.9
Germany	34	2.15	1.75	1.74	0.25	8.67	2.06	81.1
France	34	1.83	1.80	1.30	0.09	5.90	1.55	71.0
Japan	34	0.58	0.20	1.23	-1.33	3.27	0.93	212.7
China	34	3.76	2.29	5.27	-1.40	24.28	2.55	140.1
India	34	7.06	6.41	2.92	3.41	13.48	0.54	41.3
Mexico	34	9.36	5.21	8.86	2.72	35.06	1.83	94.6
South Africa	34	6.82	5.90	3.13	1.39	15.08	1.20	45.9
Nigeria	28	13.20	12.39	5.43	5.40	29.29	1.15	41.1
Egypt	34	10.31	9.20	6.13	2.26	24.39	0.88	59.5
Türkiye	34	37.06	17.96	31.87	6.25	104.54	0.56	86.0
Argentina	26	22.73	10.76	29.09	-1.17	133.49	2.56	128.0
Brazil	34	252.5	6.74	688.5	3.20	2947.7	2.99	272.7
Russia	31	57.84	11.65	164.1	2.88	874.3	4.51	283.7

Source: IMF WEO April 2025. N = number of annual observations. SD = standard deviation. CV% = coefficient of variation. Green/red cells in stationarity table indicate I(0)/I(1) classification.

The six modelled countries span a wide inflation spectrum. The United States exhibits the narrowest distribution (mean 2.69%, SD 1.51 pp); Japan has the most unusual profile (mean 0.58%, min −1.33%), reflecting persistent post-bubble deflation. Germany's mean of 2.15% with a coefficient of variation of 81.1% is the highest among advanced economies, driven by post-reunification adjustment and the 2022 energy shock. India is the most consistently elevated at mean 7.06%, SD 2.92 pp, and the lowest CV (41.3%) among emerging economies, indicating relatively stable (if high) inflation. China's large SD of 5.27 pp reflects its dramatic transition-era price liberalization of the early 1990s.

Figure 1 below presents country-level time series. To preserve comparability, the 15 economies are grouped into four tiers: advanced economies share one linear axis (max < 10%); moderate emerging markets share another (max < 35%); high-inflation economies (Türkiye, Argentina) are plotted on a capped linear scale; and hyperinflationary economies (Brazil, Russia) require a $\log_{10}$ axis.

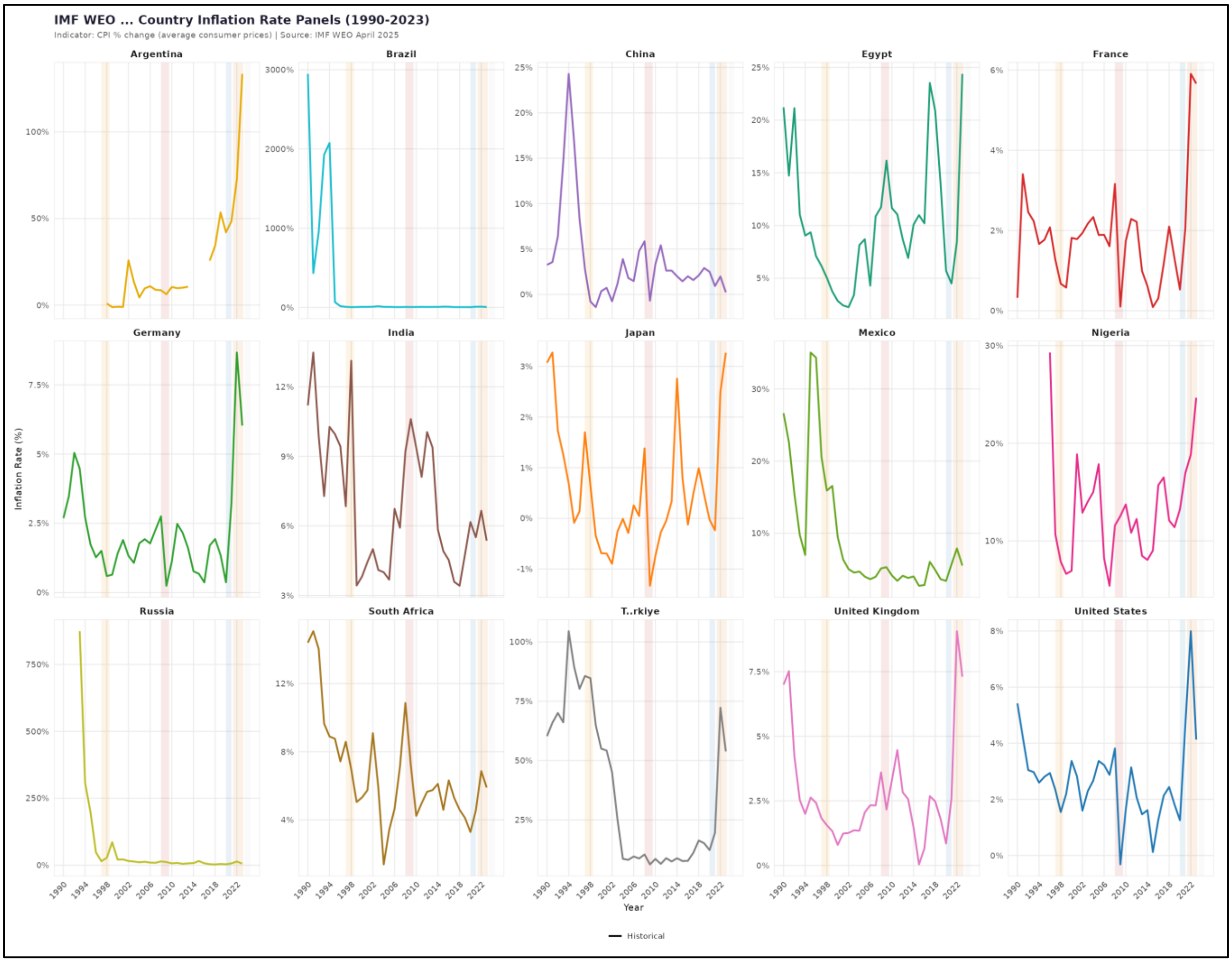

Figure 1: Annual Inflation Panels (1990–2023) — All 15 Economies. Shaded bands denote the 1997 Asian crisis, 2008 GFC, and 2020/2022 COVID events.

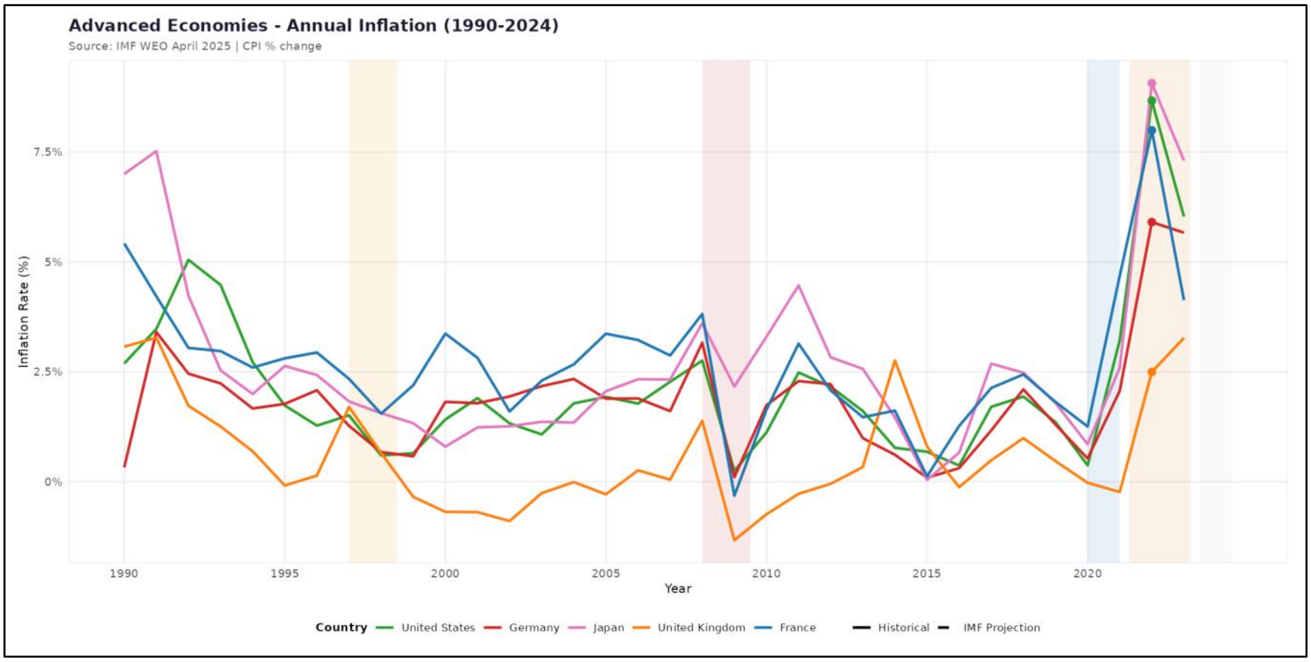

Figure 2: Advanced Economies (USA, GBR, DEU, FRA, JPN) — Annual Inflation (1990–2024).

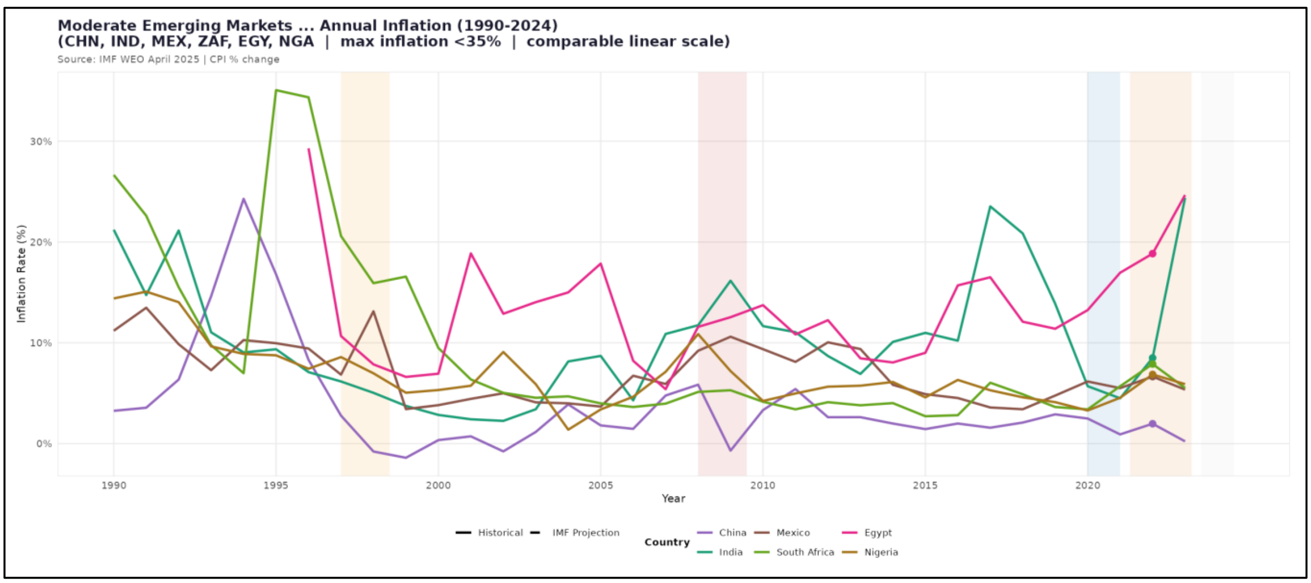

Figure 3: Moderate Emerging Markets (CHN, IND, MEX, ZAF, EGY, NGA) — Annual Inflation (1990–2024). Y-axis capped at 35% for comparability.

4. Hypothesis Testing and Results

4.1 Stationarity Tests

H_0	Inflation series is non-stationary (has a unit root) - $\gamma = 0$
H_1	Inflation series is stationary - $\gamma < 0$

ADF Regression Equation

$$\Delta y_t = \alpha + \gamma\, y_{t-1} + \delta_1 \Delta y_{t-1} + \cdots + \delta_p \Delta y_{t-p} + \varepsilon_t \quad \textit{(ADF test regression)}$$

where Δy_t = first difference of inflation, γ = coefficient on lagged level (test parameter), ε_t = white noise error. The null hypothesis is rejected when $\gamma < 0$ with a sufficiently small p-value (< 0.05).

Table 2 reports joint stationarity test results for all fifteen economies. The ADF test fails to reject the unit-root null for all six modelled countries at 5% level — (p-values: 0.081 (China) to 0.892 (USA)). PP test corroborates this uniformly. The KPSS test fails to reject the stationarity null for all six modelled countries (all p-values at the 0.10 boundary), technically contradicting ADF and PP. The decision to apply first differencing (d = 1) is defensible on the weight of evidence from ADF/PP, and slow ACF decay in the raw series.

Table 2: Stationarity Test Results — All 15 Economies

ISO	Country	ADF Stat	ADF p	PP Stat	PP p	KPSS Stat	KPSS p	Stationary
USA	United States	-1.178	0.892	-16.66	0.087	0.149	0.100	No
GBR	United Kingdom	-2.131	0.522	-7.984	0.620	0.143	0.100	No
DEU	Germany	-1.373	0.816	-9.039	0.551	0.151	0.100	No
FRA	France	-0.596	0.969	-16.16	0.096	0.130	0.100	No
JPN	Japan	-2.326	0.447	-9.897	0.496	0.199	0.100	No
CHN	China	-3.355	0.081	-12.88	0.302	0.289	0.100	No
IND	India	-1.828	0.640	-16.88	0.082	0.319	0.100	No
BRA	Brazil	-4.674	0.010	-23.94	0.010	0.528	0.035	**Yes**
RUS	Russia	-5.594	0.010	-19.56	0.035	0.480	0.046	**Yes**

ISO	Country	ADF Stat	ADF p	PP Stat	PP p	KPSS Stat	KPSS p	Stationary
TUR	Türkiye	-0.503	0.976	-3.496	0.908	0.556	0.029	No
ARG	Argentina	1.812	0.990	12.73	0.990	0.715	0.012	No
MEX	Mexico	-1.826	0.640	-13.61	0.254	0.641	0.019	No
ZAF	South Africa	-2.934	0.211	-9.915	0.495	0.570	0.026	No
NGA	Nigeria	-1.882	0.618	-22.02	0.014	0.162	0.100	No
EGY	Egypt	-2.643	0.324	-11.65	0.382	0.197	0.100	No

Green = stationary (I(0)), Red = non-stationary (I(1)). ADF/PP null: unit root. KPSS null: stationarity.

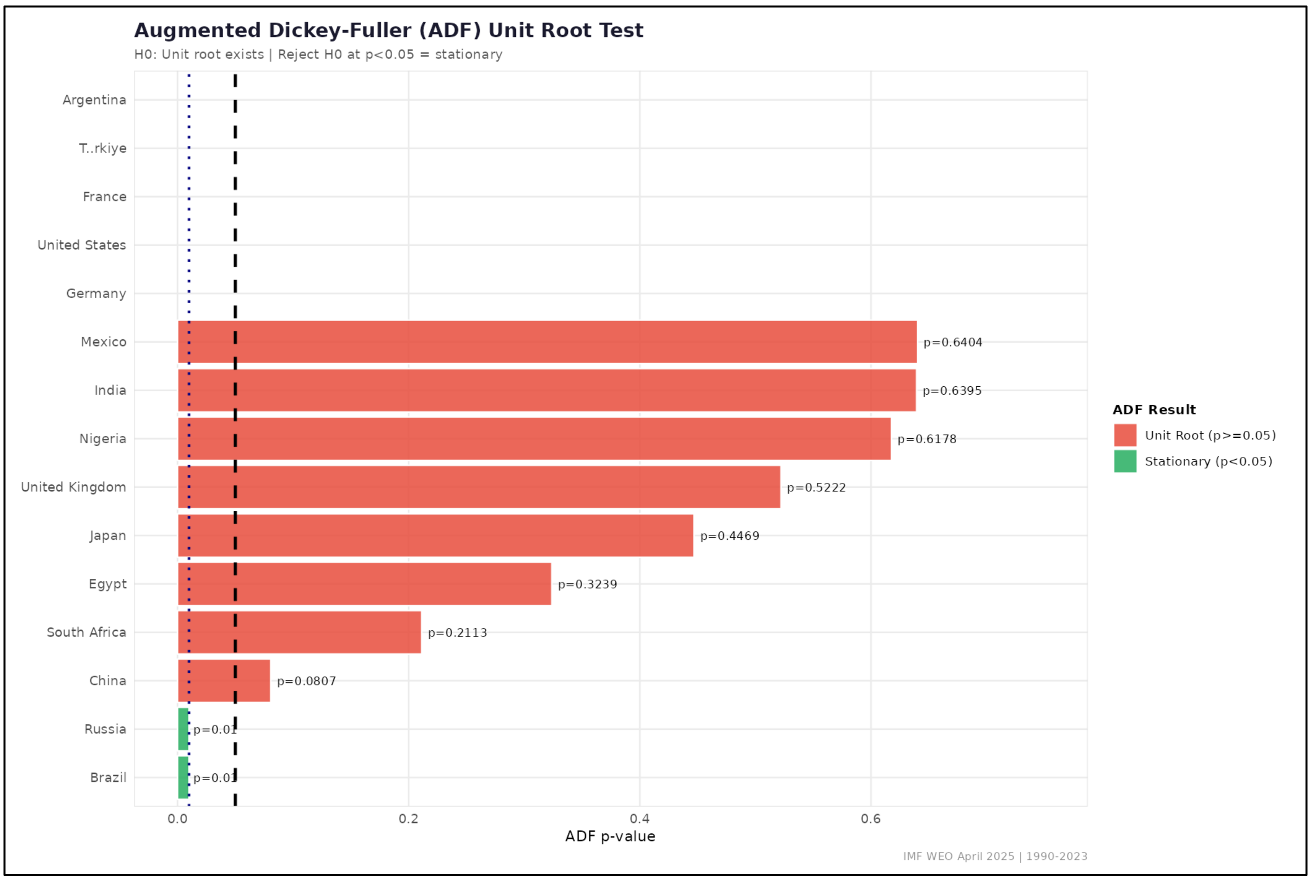

Figure 4: ADF p-values by country. Values above the dashed line (0.05) indicate failure to reject the unit-root null.

4.2 ARIMA Model Selection

Table 3 reports the best-performing ARIMA specification per country based on AICc, alongside pre-COVID out-of-sample RMSE. All six countries select first-differenced models, confirming d = 1 for the modelling sample. The United States and India select ARIMA(0,1,1), Germany selects ARIMA(0,1,2) — the only country with a second MA term — and Japan and the UK both select the parsimonious ARIMA(0,1,0) (random walk with drift), consistent with their near-unit-root dynamics. China also selects ARIMA(0,1,1). The narrow spread of AICc values within each country's grid indicates that no single specification strongly dominates, a pattern typical of short annual series with modest autocorrelation structure.

Table 3: ARIMA Model Selection — Best Specification per Country (AICc, Train 1990–2014)

Country	Best Model	AICc	AIC	BIC	RMSE (pre-COVID)	MAE (pre-COVID)
United States	ARIMA(0,1,1)	76.57	76.03	78.38	0.863	0.642
Germany	ARIMA(0,1,2)	66.59	65.45	68.98	0.553	0.449
Japan	ARIMA(0,1,0)	71.25	71.08	72.26	2.266	2.234
United Kingdom	ARIMA(0,1,0)	72.72	72.54	73.72	1.029	0.958
China	ARIMA(0,1,1)	138.24	137.70	140.05	0.808	0.624
India	ARIMA(0,1,1)	120.08	119.53	121.89	1.708	1.592

AICc = bias-corrected Akaike criterion used for model selection. RMSE and MAE are out-of-sample on pre-COVID test window (2015–2019).

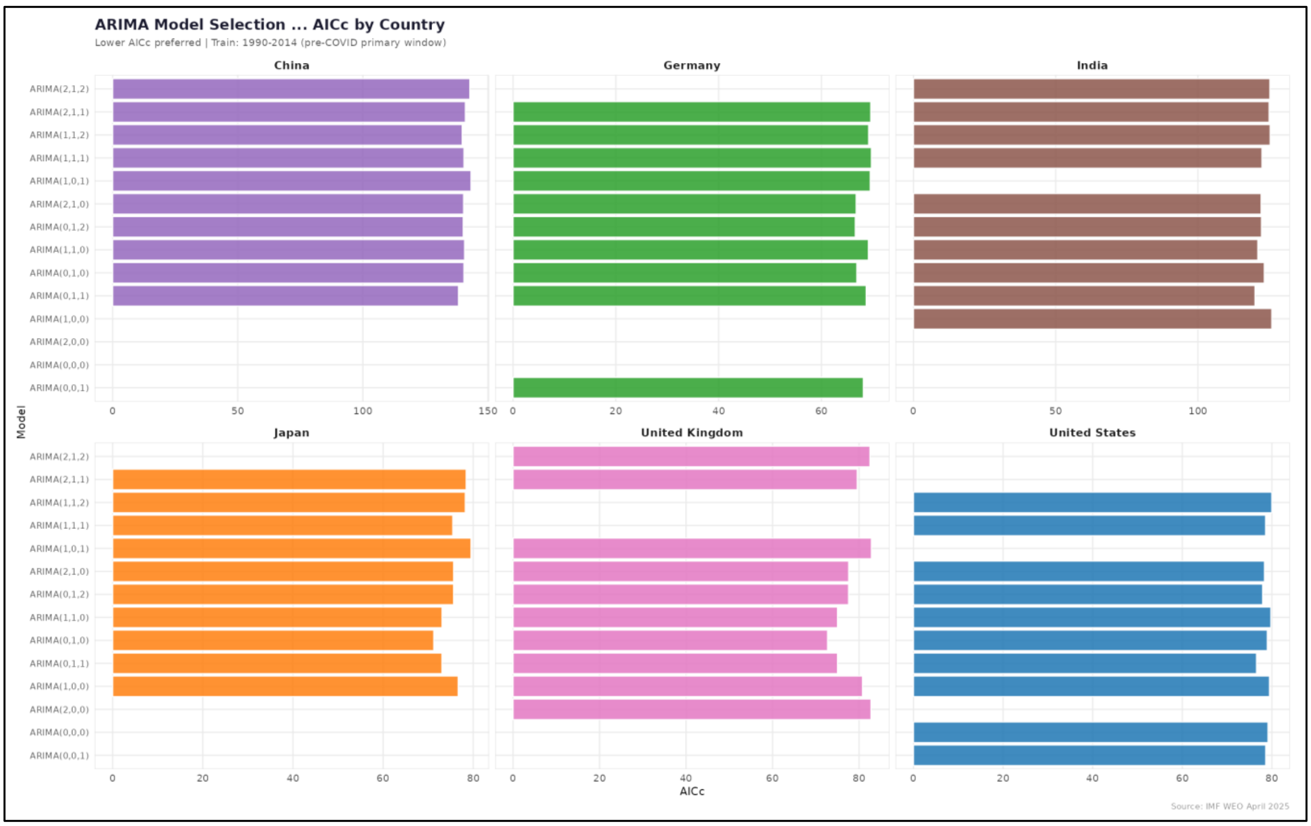

Figure 5: AICc values across all grid-search candidates by country. Lower AICc preferred. Narrow range indicates limited discriminating power of training data.

4.3 Residual Diagnostics and ARCH Tests

H_0	No ARCH effect — variance is constant (homoscedastic)
H_1	ARCH effect exists — variance is time-varying (heteroscedastic)

ARCH(1) Test Auxiliary Regression

$$\varepsilon_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + u_t \quad \textit{(Engle 1982)}$$

where ε_t^2 = squared residuals from the mean equation. A significant α_1 rejects H_0, confirming the presence of volatility clustering and motivating GARCH modelling.

Table 4 presents residual adequacy tests. The Ljung-Box test passes for all six countries — p-values range from 0.18 (India) to 0.98 (China) — confirming the ARIMA models have captured all identifiable serial correlation. However, the Shapiro-Wilk normality test rejects Gaussianity for all six countries (SW p-values - <0.001 (Germany, UK) to 0.035 (USA), with excess kurtosis particularly severe for the UK (6.67) and Germany (3.51). This implies that standard confidence intervals on forecasts are systematically too narrow, underestimating tail risk. The ARCH-LM test finds no significant volatility clustering for five of six countries; China is the marginal exception (ARCH(5) p = 0.079), consistent with its volatile early-transition inflation dynamics.

Table 4: Residual Diagnostics and ARCH Test Results

Country	Resid I(0)	LB Test	SW Normal	LB p	SW p	Skew	Ex.Kurt	ARCH5 p	ARCH10 p	ARCH?
United States	Yes	Yes (good)	No	0.480	0.035	-0.311	1.337	0.149	0.518	No
Germany	No	Yes (good)	No	0.910	0.001	1.530	3.507	0.855	0.995	No
Japan	No	Yes (good)	No	0.955	0.020	0.552	0.410	0.388	0.746	No
United Kingdom	No	Yes (good)	No	0.676	0.000	2.152	6.665	1.000	1.000	No
China	No	Yes (good)	No	0.983	0.001	1.353	2.557	0.079	0.344	Yes
India	No	Yes (good)	No	0.176	0.019	0.048	1.454	0.211	0.515	No

LB = Ljung-Box test (H_0: no autocorrelation). SW = Shapiro-Wilk (H_0: normality). ARCH5/10 = ARCH-LM at lags 5 and 10. Green = good, Red = concern.

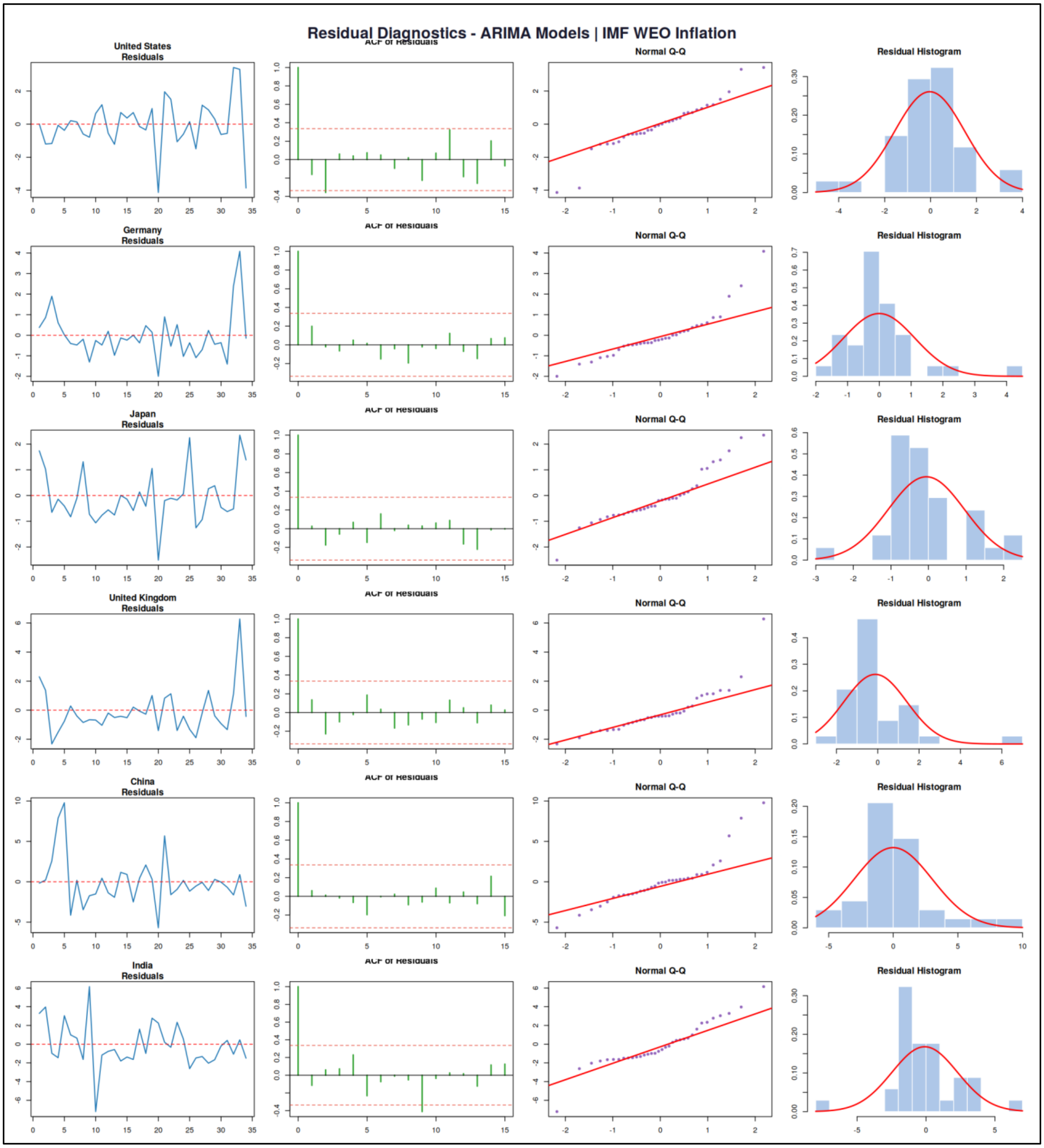

Figure 6: Residual Diagnostics — ARIMA Residuals, ACF, QQ Plots, and Histograms for all 6 Countries. QQ plot deviations confirm non-Gaussianity.

4.4 GARCH Results

Table 5 summarizes the best-fitting GARCH model per country. For four countries (USA, Germany, UK, China), the estimated α_1 = 1.0 with β_1 = 0 - a boundary solution where the optimizer cannot locate an interior maximum. This is a well-known consequence of fitting GARCH on 33–34 annual observations. India is the notable exception: GARCH(1,1)-N yields α_1 = 0.050, β_1 = 0.919, and persistence $\alpha_1 + \beta_1$ = 0.969 — an economically plausible estimate suggesting that inflation volatility shocks in India dissipate slowly. Japan's GARCH(1,1)-N yields a non-boundary α_1 = 0.537, though the β_1 = 0 result reduces it effectively to an ARCH(1) structure.

Table 5: GARCH Model Results — Best Specification per Country

Country	Best Model	AIC	BIC	ω (omega)	α_1	β_1	Persistence	Interpretation
United States	GARCH(1,1)-t	3.594	3.821	0.880	1.000	0.000	1.000	Boundary
Germany	GARCH(1,1)-t	3.395	3.619	0.408	1.000	0.000	1.000	Boundary
Japan	GARCH(1,1)-N	3.319	3.499	0.794	0.537	0.000	0.537	Partial
United Kingdom	GARCH(1,1)-t	3.928	4.153	1.562	1.000	0.000	1.000	Boundary
China	GARCH(1,1)-N	5.036	5.215	1.164	1.000	0.071	1.071	Explosive
India	GARCH(1,1)-N	5.139	5.318	0.000	0.050	0.919	0.969	Stable

Persistence = α_1 + β_1. Values = 1.0 indicate boundary solutions. India is the only country yielding a statistically meaningful GARCH estimate.

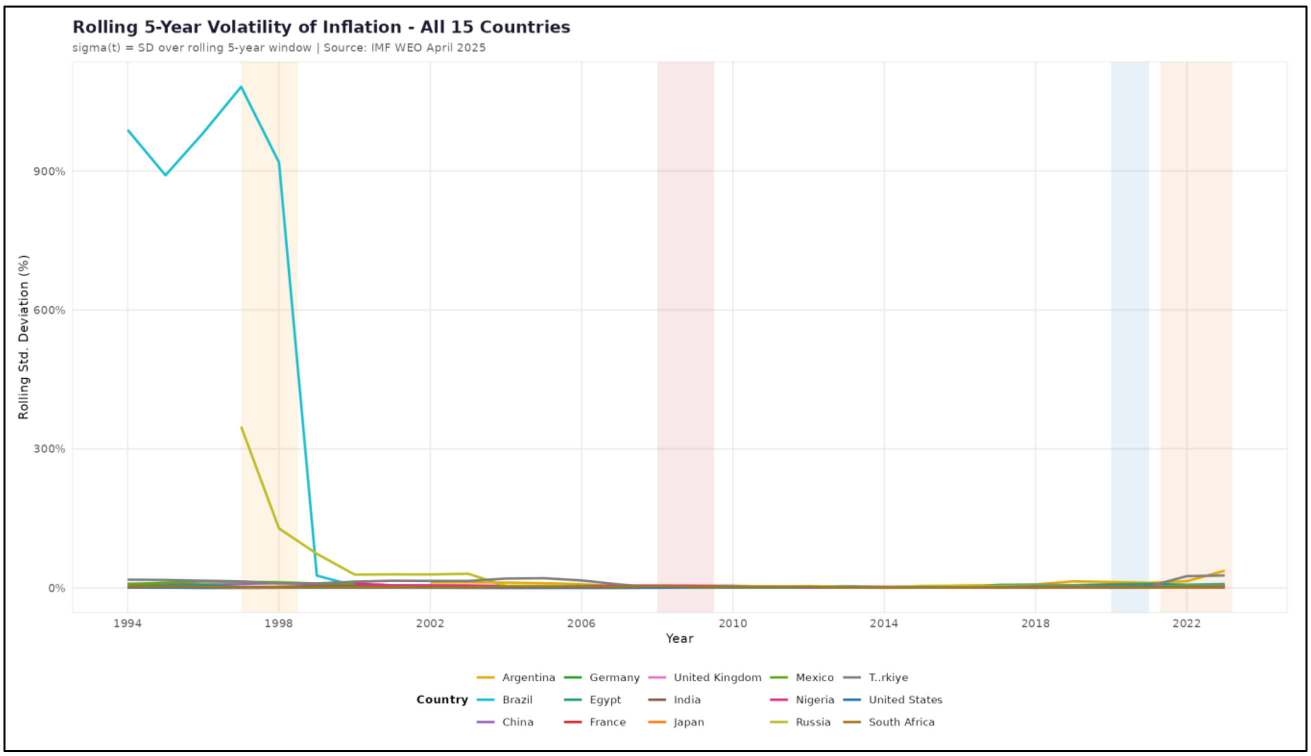

Figure 7: Rolling 5-Year Inflation Volatility (Standard Deviation) — All 15 Countries (1994–2023). Note the dramatic decline in BRA and RUS volatility post-stabilisation.

5. Findings and Interpretation

5.1 Primary Forecast Performance (Pre-COVID, 2015–2019)

Table 6 presents the complete forecast accuracy results across both evaluation windows. In the pre-COVID primary window, Japan is the standout performer: RMSE(0.383 pp) against a naive benchmark (2.266 pp) yields a skill score of 83.1% — by far the highest in the sample. Japan's ARIMA(0,1,0) model exploits the strong mean-reverting structure of near-zero inflation: once the level is established, the series has high persistence. Germany is the second most skillful (skill score 32.6%, RMSE 0.499 pp vs. naive 0.741 pp), suggesting a detectable moving-average structure in its ARIMA(0,1,2) specification.

In sharp contrast, the United States (skill score 0.5%, RMSE 0.813 pp) and China (5.5%, RMSE 0.485 pp) deliver near-zero improvement over a random walk. The USA result has a clear economic interpretation: US inflation in 2015–2019 evolved in a manner indistinguishable from a random walk, and ARIMA adds essentially no exploitable predictive content. India's pre-COVID skill score of exactly 0.0% — ARIMA RMSE(1.708 pp) identical to the naive — indicates that the random walk is a precise match for ARIMA accuracy over this period. The UK records a modest 9.4% skill score (RMSE 0.933 pp vs. naive 1.029 pp).

Table 6: Forecast Accuracy Metrics — Pre-COVID Primary and COVID-Era Stress Test

Country	Window	RMSE	MAE	sMAPE%	Naive RMSE	Skill Score	Bias
United States	Pre-COVID	0.813	0.681	55.8	0.817	0.005	0.03
Germany	Pre-COVID	0.499	0.402	38.5	0.741	0.326	0.10
Japan	Pre-COVID	0.383	0.295	66.1	2.266	0.831	-0.06
United Kingdom	Pre-COVID	0.933	0.827	70.0	1.029	0.094	0.49
China	Pre-COVID	0.485	0.377	18.5	0.513	0.055	-0.09
India	Pre-COVID	1.708	1.592	32.5	1.708	0.000	1.59
United States	COVID-era	3.118	2.569	67.5	3.608	0.136	-2.12
Germany	COVID-era	4.113	3.385	103.3	4.465	0.079	-2.89
Japan	COVID-era	1.541	1.409	151.3	1.782	0.135	-0.56
United Kingdom	COVID-era	4.146	3.231	76.9	4.605	0.100	-2.66
China	COVID-era	1.637	1.422	80.6	1.743	0.061	1.42
India	COVID-era	0.729	0.668	11.3	1.264	0.423	-0.00

Green (dark) = skill score > 30%. Green (light) = positive but low. Red = near-zero or negative. Yellow rows = COVID-era window. Bias = mean error (negative = under-prediction).

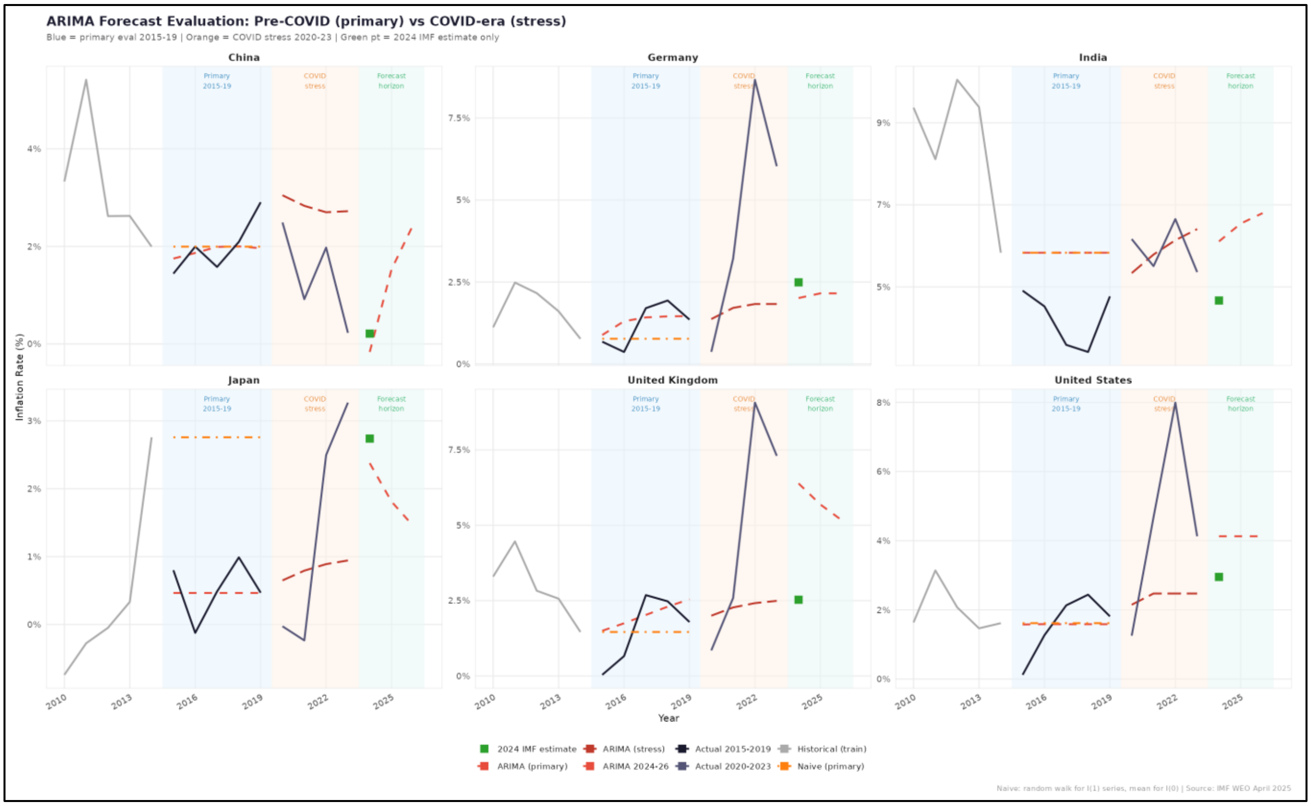

Figure 8: Forecast Evaluation — ARIMA vs Naive Benchmark, Both Windows. Blue shading = pre-COVID primary evaluation (2015–2019). Orange shading = COVID stress (2020–2023). Green square = 2024 IMF near-final estimate.

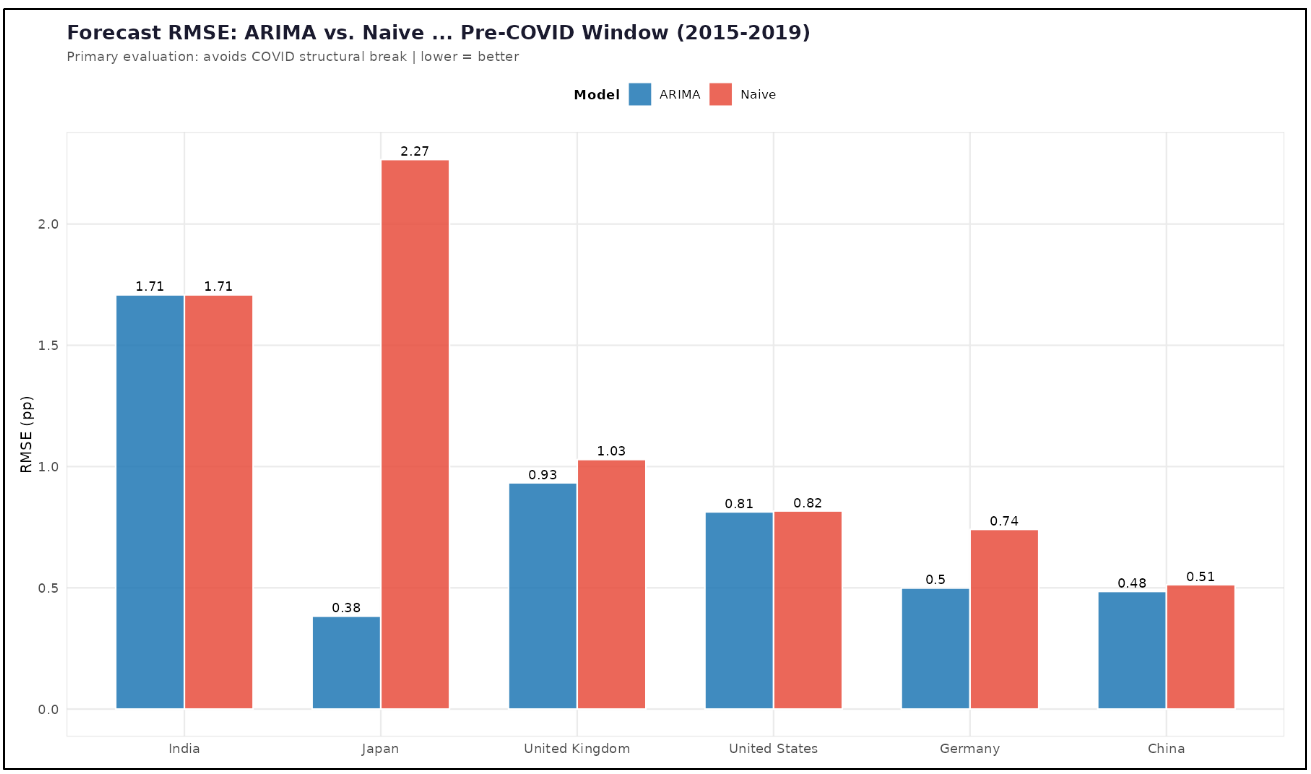

Figure 9: RMSE Comparison — ARIMA vs Naive Benchmark (Pre-COVID 2015–2019). Lower RMSE = better forecast accuracy.

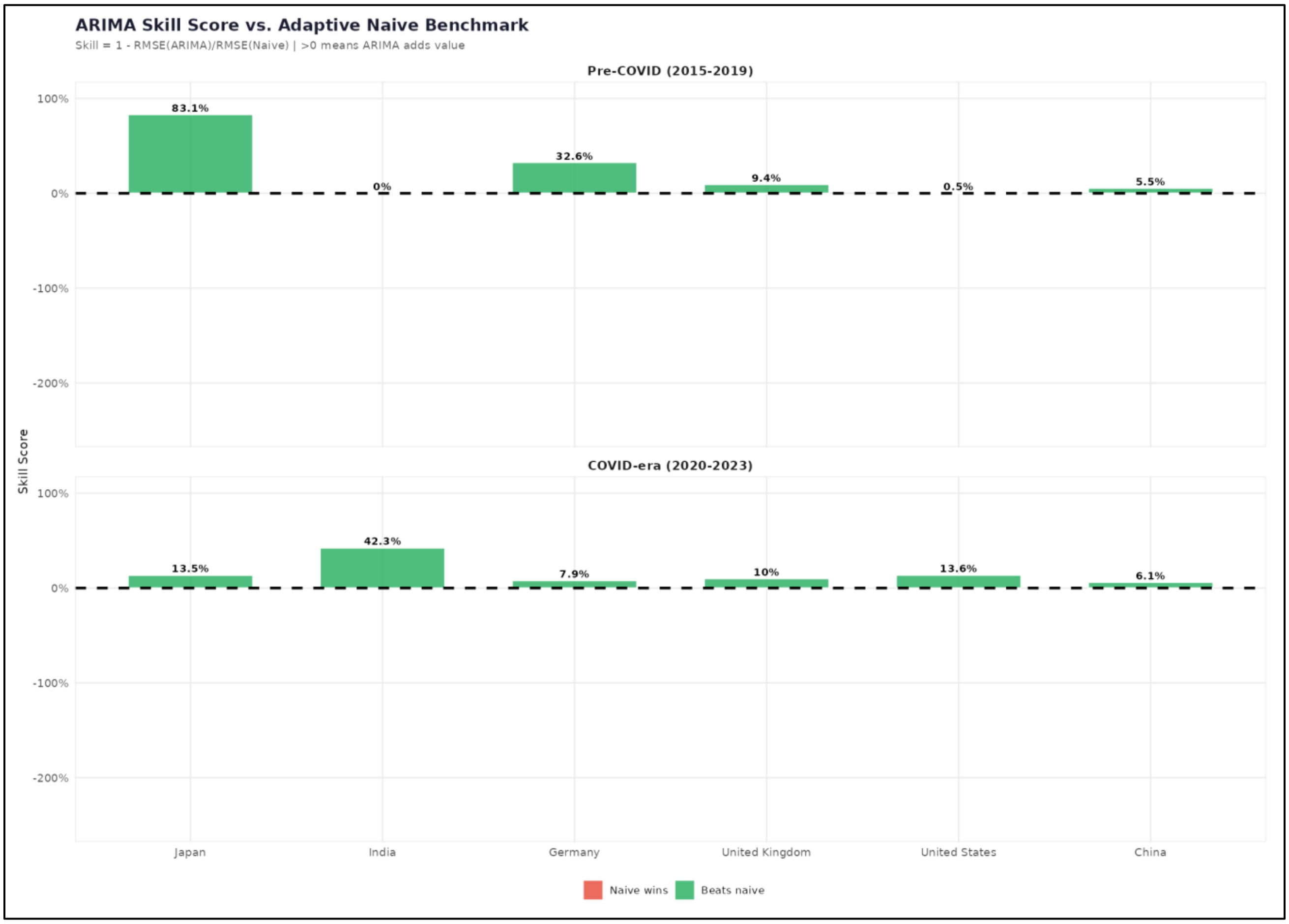

Figure 10: ARIMA Skill Score vs Adaptive Naive Benchmark — Pre-COVID (top) and COVID-era (bottom). Positive scores indicate ARIMA beats the random walk.

5.2 COVID-Era Stress Test (2020–2023)

The COVID-era evaluation confirms expected accuracy deterioration across all economies, though the pattern of relative performance is informative. Germany records an RMSE of 4.113 pp, reflecting the severity of the 2022 energy-price surge to 8.7% — far outside the 1990–2019 training distribution. The UK RMSE of 4.146 pp is the highest in the sample, with a bias of −2.655 pp confirming systematic under-prediction. India records the best COVID-era performance (RMSE 0.729 pp, skill score 42.3%) because its inflation remained contained within a historically familiar range during 2020–2023, allows ARIMA model's mean-reverting forecast to remain accurate. All advanced economies show negative bias (under-prediction) during the COVID window, reflecting that ARIMA models calibrated on low-inflation training data have no mechanism to anticipate supply-chain, fiscal, and energy shocks.

5.3 The sMAPE Anomaly and Metric Choice

sMAPE figures require careful interpretation. Japan records a pre-COVID sMAPE of 66.1% despite havi RMSE (0.383 pp) in the sample. Japan's COVID-era sMAPE of 151.3% is similarly inflated. Where inflation regimes differ markedly across countries, RMSE and the skill score are the more appropriate primary ranking metrics, not sMAPE.

5.4 ARIMA vs IMF 2024 Estimate

Table 7 compares the ARIMA 2024 forecast against the IMF near-final estimate. Germany (−0.48 pp), Japan (−0.36 pp), and China (−0.38 pp) are within half a percentage point, representing acceptable proximity. The UK divergence of +3.86 pp is the most concerning: the ARIMA model, anchored by the 2022–2023 inflation spike of 9.1% and 7.3%, mean-reverts far too slowly and projects 6.39% for 2024 against the IMF's 2.53%. India's +1.44 pp over-prediction reflects similar slow reversion following its own post-COVID surge. These divergences illustrate the structural limitation of univariate models in capturing policy-driven disinflation.

Table 7: ARIMA 2024 Forecast vs IMF Near-Final Estimate

Country	ARIMA 2024 (%)	IMF Estimate (%)	Difference (pp)	Direction
United States	4.13	2.95	+1.18	Over
Germany	2.01	2.49	-0.48	Under
Japan	2.38	2.74	-0.36	Under
United Kingdom	6.39	2.53	+3.86	Over
China	-0.16	0.21	-0.38	Under
India	6.11	4.67	+1.44	Over

Green = ARIMA under-predicts relative to IMF. Red = ARIMA over-predicts. 2024 IMF values are near-final estimates, not finalized outturns.

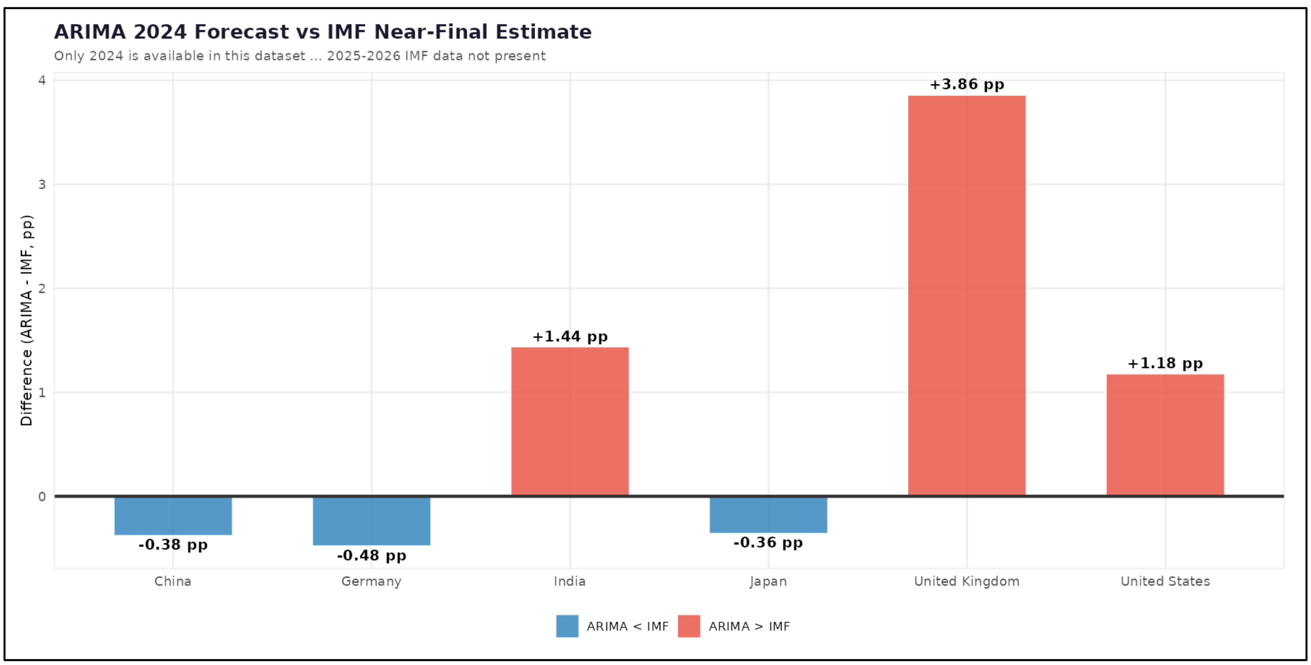

Figure 11: ARIMA 2024 Forecast vs IMF Estimate — Divergence in Percentage Points. Positive = ARIMA over-predicts; Negative = under-predicts.

Figure 12: ARIMA Forecasts 2024–2026 with 80% and 95% Confidence Intervals. Green square = 2024 IMF near-final estimate. Red triangles = ARIMA point forecasts.

5.5 Economic Interpretation

The overarching finding is that inflation in most modelled economies behaves sufficiently like a random walk that a univariate ARIMA model struggles to deliver consistent, economically meaningful forecast improvements. This aligns with Atkeson and Ohanian's (2001) finding that models without external information rarely beat naive benchmarks. Germany and Japan are partial exceptions: Germany's moving-average structure is detectable in-sample, while Japan's near-zero inflation regime creates a form of predictability that ARIMA exploits well. The GARCH results reinforce the data-frequency limitation: with the exception of India (persistence 0.969 — a plausible estimate of slow shock dissipation), all other GARCH solutions degenerate to boundary estimates, confirming that annual data cannot sustain reliable volatility modelling.

6. Conclusion

6.1 Summary

This study implemented a complete Box-Jenkins ARIMA pipeline on annual CPI inflation series for six economies from the IMF WEO database. A dual-window evaluation design — pre-COVID primary validation (2015–2019) and COVID-era stress testing (2020–2023) — provides a more honest assessment of forecast reliability than a single window. ARIMA delivers meaningful predictive skill only for economies with strong autocorrelation structure and stable regimes: Japan (83.1%) and Germany (32.6%). For USA, China, and India under pre-COVID conditions, improvement over a random walk is negligible. Ljung-Box tests confirm model adequacy but reveals non-Gaussian errors, implying underestimated forecast uncertainty. GARCH estimation is severely constrained by the 34-observation annual sample, yielding reliable results for India only.

6.2 Limitations

The most fundamental limitation is the univariate nature of the models: excluding output gaps, policy rates, and commodity prices causing systematic misses where these drivers dominate. Annual data frequency limits unit-root test power and GARCH reliability. The COVID structural break is handled through separate evaluation windows rather than formal intervention modelling.

6.3 Future Directions

Future work should explore ARIMAX models with leading macroeconomic indicators as exogenous regressors and Vector Autoregression (VAR) models to capture cross-country inflation spillovers. Monthly or quarterly data would dramatically increase test power, enable seasonal ARIMA, and render GARCH estimation meaningful. Regime-switching models such as Markov-switching ARIMA - could explicitly accommodate the distinct inflation regimes visible in the China and India series.

7. References

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